



C2 - Machine Learning Project

C-DAT-300

Sentiment Analysis

in real-time using Twitter

1.1



Sentiment Analysis

delivery method: endtoendml on Github
language: python



- The totality of your source files, except all useless files (binary, temp files, obj files,...), must be included in your delivery.

WHAT IS KAFKA

Event streaming is the practice of capturing data in real-time from event sources like databases, sensors, mobile devices, cloud services, and software applications in the form of streams of events; storing these event streams durably for later retrieval; manipulating, processing, and reacting to the event streams in real-time as well as retrospectively; and routing the event streams to different destination technologies as needed. Event streaming thus ensures a continuous flow and interpretation of data so that the right information is at the right place, at the right time.

EVENTS

An **event** records the fact that “something happened” in the world or in your business. It is also called record or message in the documentation. When you read or write data to Kafka, you do this in the form of events. Conceptually, an event has a key, value, timestamp, and optional metadata headers. Here’s an example event:

```
Event key: "Alice"  
Event value: "Made a payment of $200 to Bob"  
Event timestamp: "Jun. 25, 2020 at 2:06 p.m."
```

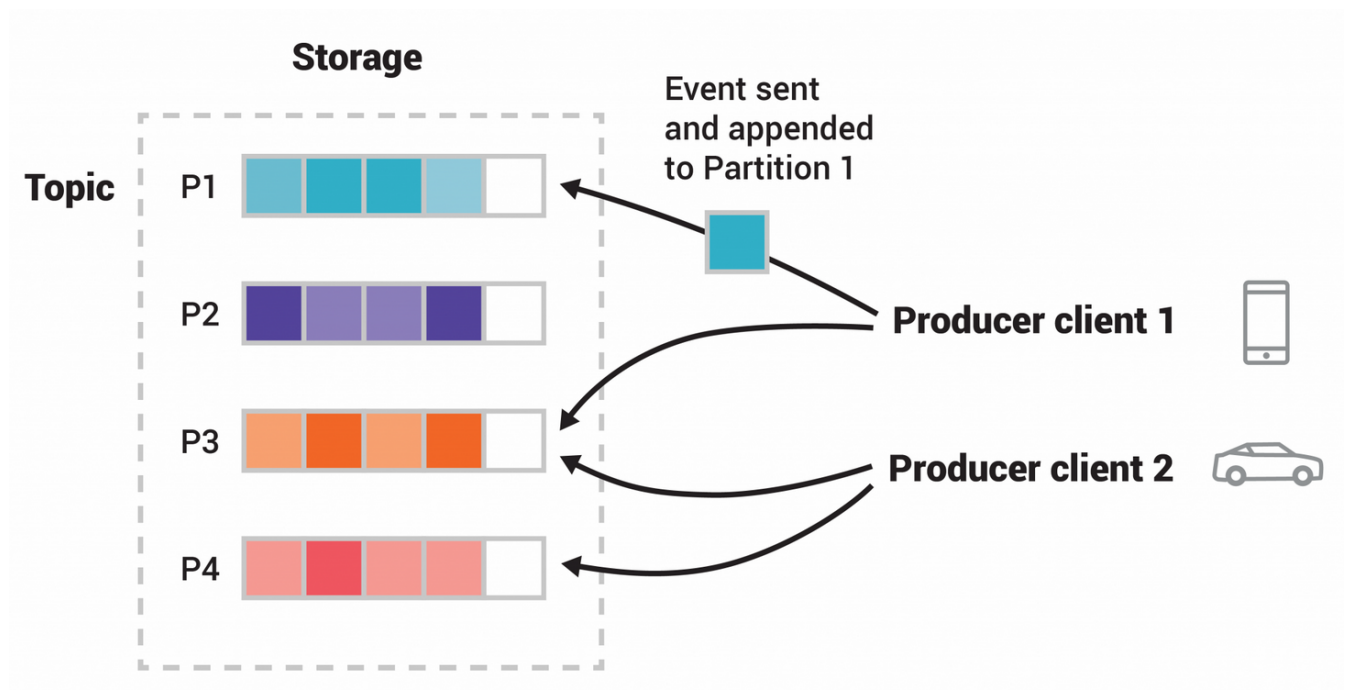
PRODUCERS / CONSUMERS

Producers are those client applications that **publish** (write) events to Kafka, and consumers are those that **subscribe to** (read and process) these events. In Kafka, producers and consumers are fully decoupled and agnostic of each other, which is a key design element to achieve the high scalability that Kafka is known for. For example, producers never need to wait for consumers.



TOPICS

Events are organized and durably stored in **topics**. Topics in Kafka are always multi-producer and multi-subscriber: a topic can have zero, one, or many producers that write events to it, as well as zero, one, or many consumers that subscribe to these events. Topics in Kafka are always multi-producer and multi-subscriber: a topic can have zero, one, or many producers that write events to it, as well as zero, one, or many consumers that subscribe to these events. Events in a topic can be read as often as needed, you define for how long Kafka should retain your events through a per-topic configuration setting, after which old events will be discarded.





PROJECT SUMMARY

The goal of this project is to create an end-to-end Machine Learning project, including :

- **extract** tweets of specifics topics from Twitter, in real-time using Apache Kafka
 - **transform**, using you trained-model for sentiments analysis classification
 - **load** data into a data-warehouse using PostgreSQL
 - **real-time dashboard**, to monitor the results for each topics using PowerBI (or equivalent)
- Each parts can be start independently.



EXTRACT

First you need to install Kafka (and Zookeeper to manage it) on your system. At least 4GB of RAM is needed.

You can use this tutorial : <https://www.digitalocean.com/community/tutorials/how-to-install-apache-kafka-on-ubuntu-18-04>

You need access to Twitter API. Create of project on the developer interface of Twitter to get your Acces Token : <https://developer.twitter.com/en/portal/projects-and-apps>

Using python (**tweepy** package for Twitter API and **kafka-python** for Kafka) extract Tweets of differents Netflix shows (The Queen's Gambit and The Crown for exemple) and publish them into the Kafka server using topics. You can have multiple #keyword to monitor for one specific show.

TRANSFORM

First you need to create a **Sentiment Analysis** model with the IMDB database using Scikit-learn (or XG-Boost).

Download the Dataset : <https://epitechfr.sharepoint.com/:f:/s/Peda/EtTNIWnuEd5MgPY1Q-3MP34BbvIt-wRUxUQJqXO>

Then, using python, create a Kafka "consumer" with your trained-model to classify each tweets from Kafka.

LOAD

Make another Kafka consumer to export tweets and their label inside a PostgreSQL.



REAL-TIME DASHBOARD

Create one PowerBI dashboard connected on the Kafka in order to monitor in real-time the number of tweets coming by topics

Create another PowerBI connected to the PostgreSQL database to monitor the results of your classifier.